

Product Requirements Document (PRD)

Siri Dhanya Green Hub

1. Project Overview

Project Name: Siri Dhanya Green Hub

Platform: Android Application

Domain: Agriculture / Millet Marketplace / Farmer Support System

Siri Dhanya Green Hub is a smart agriculture mobile application designed to support millet farmers by providing market prices, crop information, mandi updates, analytics graphs, and farmer-friendly digital services in one platform.

The application aims to improve millet awareness, help farmers make informed selling decisions, and digitally connect users with agricultural information.

2. Problem Statement

Many millet farmers face difficulties such as:

- Lack of real-time mandi price information
- Limited awareness about millet benefits
- Poor access to organized agricultural market data
- Difficulty tracking crop trends and prices
- Limited digital platforms focused on millet ecosystems

Because of this, farmers often sell products at lower prices and lack proper market insights.

3. Proposed Solution

Siri Dhanya Green Hub provides:

- Live mandi/market price updates
- Millet information and categories
- Graphical price analysis
- Farmer-friendly UI
- Authentication system for secure access
- Digital agricultural awareness platform

The system helps farmers access important information quickly and make better agricultural and selling decisions.

4. Objectives

Primary Objectives

- Support millet farmers digitally
- Provide mandi price transparency
- Promote millet awareness
- Simplify agricultural market tracking

Secondary Objectives

- Improve user engagement with agriculture technology
- Encourage smart farming decisions
- Build a scalable agriculture platform

5. Target Users

User Type	Description
Farmers	View mandi prices and crop details
Consumers	Learn about millet products
Agricultural Students	Access agricultural information
Vendors/Traders	Monitor market trends

6. Core Features

6.1 User Authentication

Users can:

- Register account
- Login securely
- Access personalized dashboard

Technology Used

- Firebase Authentication

6.2 Home Dashboard

The dashboard displays:

- Millet categories
- Quick navigation cards
- Market highlights
- Agricultural information

6.3 Millet Information Module

Users can:

- Browse different millet types
- View nutritional and farming information
- Learn millet benefits

Example Millets

- Ragi
- Foxtail Millet
- Jowar
- Bajra
- Little Millet

6.4 Mandi Price Screen

Displays:

- Current market prices
- Crop rates
- Location-wise pricing
- Market trends

Benefits

- Helps farmers choose better selling times
- Improves market awareness

6.5 Graph & Analytics Module

The app includes:

- Price trend graphs
- Crop analytics
- Market comparison visualization

Purpose

- Easy understanding of market fluctuations
- Visual decision-making support

6.6 Navigation System

The app contains:

- Bottom navigation bar
- Smooth screen transitions
- Organized UI structure

Example Navigation

- Home
- Market
- Graphs
- Profile

7. Functional Requirements

ID	Requirement
FR-1	User shall register/login securely
FR-2	System shall display millet categories
FR-3	System shall show mandi prices
FR-4	System shall display graphical analytics
FR-5	User shall navigate between screens
FR-6	System shall store user authentication data

8. Non-Functional Requirements

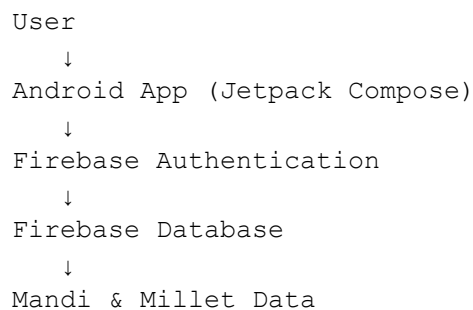
Category	Requirement
Performance	Fast screen loading
Security	Secure Firebase authentication
Usability	Farmer-friendly simple UI
Scalability	Future feature expansion support

Reliability Stable data handling

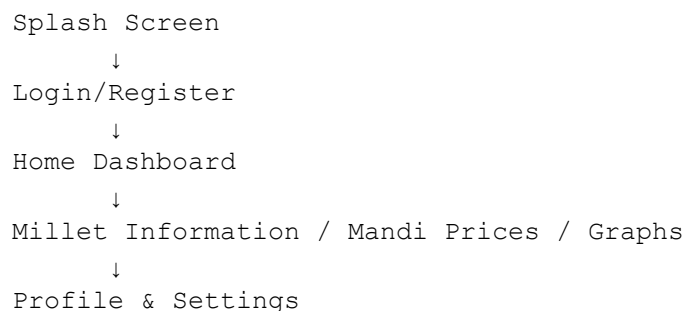
9. Technology Stack

Component	Technology
Frontend	Kotlin + Jetpack Compose
Backend	Firebase
Authentication	Firebase Authentication
Database	Firebase Realtime Database / Firestore
Charts	Compose Charts / MPAndroidChart
IDE	Android Studio

10. System Architecture



11. User Flow



12. UI Modules

Screen	Purpose
Splash Screen	App branding
Login Screen	User authentication
Home Screen	Dashboard
Market	Mandi prices

Screen

Graph Screen Analytics

Profile Screen User details

13. Future Scope

Future enhancements may include:

- AI-based crop prediction
- Weather forecasting
- Multi-language support
- Online millet marketplace
- Government scheme updates
- Farmer community chat system
- IoT integration for smart farming

14. Risks & Limitations

Risk	Impact
Internet dependency	App requires connectivity
Data accuracy	Market prices must stay updated
User adoption	Farmers may need digital awareness

15. Success Metrics

The project will be considered successful if:

- Users can access mandi prices easily
- Farmers use the platform regularly
- Market data is displayed correctly
- App navigation remains smooth and simple

16. Conclusion

Siri Dhanya Green Hub is a farmer-centered agricultural application focused on millet awareness and mandi price transparency. The application combines modern Android technologies with agricultural support features to create a useful digital platform for farmers, students, and consumers.

The project contributes toward smart agriculture initiatives and promotes digital transformation in the millet ecosystem.

